Lista 03 - Resolucao

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Questao 1

```
A = (2, -1, 3), u = (1, 2, -1), v = (-2, 1, 3)
(x, y, z) = (2, -1, 3) + h(1, 2, -1) + t(-2, 1, 3)
x = 2 + h - 2t
y = -1 + 2h + t
z = 3 - h + 3t
```

Questao 2

```
P = (-1, 4, 2), n = (3, 2, -1)
3(x - (-1)) + 2(y - 4) - (z - 2) = 0
3(x + 1) + 2(y - 4) - z + 2 = 0
3x + 3 + 2y - 8 - z + 2 = 0
3x + 2y - z - 3 = 0
```

Questao 3

1)

```
P = (2, -1, 1) + h(1, 2, 0) + t(-1, 1, 3)
x = 2 + h - t
y = -1 + 2h + t
z = 1 + 3t
```

2)

```
u = (1, 2, 0), v = (-1, 1, 3)
u x v = (2 . 3 - 0 . 1, 0 . (-1) - 1 . 3, 1 . 1 - 2 . (-1))
u x v = (6, -3, 3) = 3(2, -1, 1)
n = (2, -1, 1)
2(x - 2) - (y + 1) + (z - 1) = 0
2x - 4 - y - 1 + z - 1 = 0
2x - y + z - 6 = 0
```

Questao 4

```
AB = B - A = (0 - 2, 3 - 1, 1 - 0) = (-2, 2, 1)
AC = C - A = (1 - 2, -1 - 1, 4 - 0) = (-1, -2, 4)
AB x AC = (2 . 4 - 1 . (-2), 1 . (-1) - (-2) . 4, (-2) . (-2) - 2 . (-1))
AB x AC = (10, 7, 6)
10(x - 2) + 7(y - 1) + 6(z - 0) = 0
10x - 20 + 7y - 7 + 6z = 0
10x + 7y + 6z - 27 = 0
(x, y, z) = (2, 1, 0) + h(-2, 2, 1) + t(-1, -2, 4)
```

Questao 5

```
n1 = (2, -1, 1), n2 = (1, 2, -2)
n1 . n2 = 2 . 1 + (-1) . 2 + 1 . (-2) = -2
|n1 . n2| = 2
|n1| = raiz(6), |n2| = 3
cos(theta) = 2 / (3 raiz(6)) ~= 0,2722
theta = arccos(0,2722) ~= 74,21 graus
```